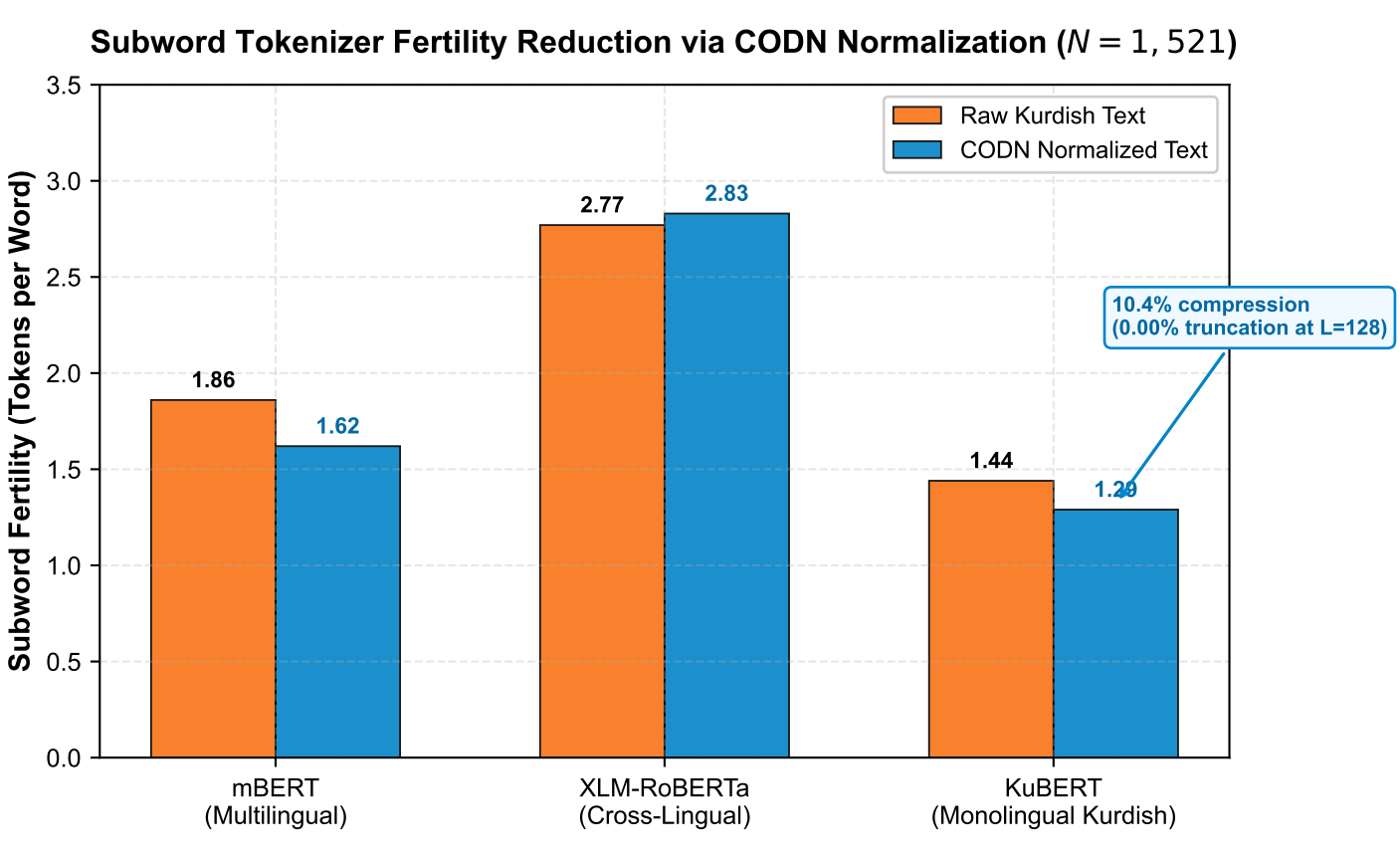
Subword Tokenizer Fertility Reduction via CODN Normalization ($N = 1,521$)
Subword Fertility (Tokens per Word)
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Raw Kurdish Text
CODN Normalized Text
1.86
1.62
2.77
2.83
1.44
1.29
10.4% compression
(0.00% truncation at L=128)
mBERT
(Multilingual)
XLM-RoBERTa
(Cross-Lingual)
KuBERT
(Monolingual Kurdish)